Star Forests and the Diagonal Sum: Proved Bounds and a Conditional Reduction

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Abstract

We give a complete proof of the diagonal upper bound for two arbitrary star forests, and of the matching lower bound when the host is required to be bipartite. We then formulate a unit-extension hypothesis and prove that, together with the assumed net-unit-excess-at-most-ten result, it implies the full diagonal formula for arbitrary hosts. We also derive a quantitative bound using the assumed base result alone. The unit-extension hypothesis is not proved here, and the supplied base-case manuscript is not independently certified. Consequently, this note is not an unconditional proof of Erdős problem 561.

1 Definitions and the desired identity

All graphs are finite and simple, unless an auxiliary multigraph is explicitly introduced. Copies are ordinary subgraphs, not necessarily induced, and every union of target stars is vertex-disjoint. Write

$$F_1 = \bigsqcup_{i=1}^{s} K_{1,n_i}, \qquad F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j},$$

where $s, t \geq 1$ and

$$n_1 \geq \cdots \geq n_s \geq 1, \qquad m_1 \geq \cdots \geq m_t \geq 1.$$

The notation $H \longrightarrow (F_1, F_2)$ means that every red–blue edge coloring of H contains a red F_1 or a blue F_2. Define

$$\widehat{R}(F_1, F_2) = \min\{e(H) : H \longrightarrow (F_1, F_2)\}.$$

Set

$$\ell_k = \max_{\substack{1 \leq i \leq s,\, 1 \leq j \leq t \\ i+j=k}} (n_i + m_j - 1), \qquad L(F_1, F_2) = \sum_{k=2}^{s+t} \ell_k.$$

Also write

$$E_1 = \sum_i n_i, \quad E_2 = \sum_j m_j, \quad c_1 = 2s - E_1, \quad c_2 = 2t - E_2, \quad c = \max(c_1, c_2).$$

Conjecture 1.1 (The unrestricted diagonal identity)**.** *For every pair of positive nonincreasing lists,*

$$\widehat{R}(F_1, F_2) = L(F_1, F_2). \tag{1}$$

The general identity is stated as Conjecture 1.1 in the July 4, 2026 version of [2]. That paper concerns uniform star forests and is not a source for the unit-extension hypothesis below.

2 The universal upper bound

Lemma 2.1 (Threshold criterion)**.** *Suppose H is a vertex-disjoint union of stars. For one fixed color, let a_h be the number of edges of that color in host component h. Its edges contain $\bigsqcup_{i=1}^{r} K_{1,d_i}$, where $d_1 \geq \cdots \geq d_r \geq 1$, if and only if, for every i, at least i values a_h are at least d_i.*

Proof. Every nonempty target star must occupy one host component. Two vertex-disjoint target stars cannot occupy the same host star, since every edge in that component meets its center. This proves necessity. Conversely, choose host components for $d_1, d_2, \ldots, d_r$ in this order. At step i, at least i components meet the threshold, and at most $i-1$ have been used. Thus a suitable unused component is available. □

Theorem 2.2 (Universal upper bound)**.** *For arbitrary positive lists, $\widehat{R}(F_1, F_2) \leq L(F_1, F_2)$.*

Proof. Let

$$H_* = \bigsqcup_{k=2}^{s+t} K_{1,\ell_k}.$$

This graph has exactly L edges. Fix any red–blue coloring and let r_k, b_k be the two color counts in component k, so $r_k + b_k = \ell_k$.

Suppose neither target occurs in its designated color. The threshold criterion supplies indices i, j for which

$$A = \{k : r_k \geq n_i\}, \quad |A| \leq i-1, \qquad B = \{k : b_k \geq m_j\}, \quad |B| \leq j-1.$$

For every integer k with $2 \leq k \leq i+j$, there exist $u \leq i$ and $v \leq j$ with $u, v \geq 1$ and $u+v = k$. Monotonicity yields

$$\ell_k \geq n_u + m_v - 1 \geq n_i + m_j - 1.$$

If $k \notin A \cup B$, then $r_k \leq n_i - 1$ and $b_k \leq m_j - 1$, which contradicts this inequality. Consequently

$$\{2, \ldots, i+j\} \subseteq A \cup B.$$

But the left side has $i+j-1$ elements, whereas $|A \cup B| \leq i+j-2$. This contradiction proves $H_* \longrightarrow (F_1, F_2)$. □

This is the upper-bound argument in Section 3 of the supplied manuscript [1]; it requires no net-unit-excess restriction.

3 An unconditional lower bound for bipartite hosts

This section is an additional derivation, not an appeal to the unverified normalization steps in [1]. For $1 \leq i \leq s$ and $1 \leq j \leq t$, put

$$F_1^{(i)} = \bigsqcup_{a=i}^{s} K_{1,n_a}, \qquad F_2^{(j)} = \bigsqcup_{b=j}^{t} K_{1,m_b}.$$

Lemma 3.1 (Deletion preserves leading-component suffixes)**.** *Suppose $H \longrightarrow (F_1, F_2)$. For every vertex set X of size q and every feasible pair i, j with $i + j = q + 2$,*

$$H - X \longrightarrow (F_1^{(i)}, F_2^{(j)}).$$

Proof. Partition X into sets X_R, X_B of sizes $i-1, j-1$. If the assertion fails, take an avoiding coloring of $H - X$. Extend it by coloring every edge meeting X_R red, and every other edge meeting X_B blue.

Every new red edge meets X_R. A red copy of F_1 has at most $i-1$ components meeting X_R, because those components are vertex-disjoint. Discard them. Every remaining component lies entirely in $H - X$: a red edge meeting X_B would also meet X_R, by the definition of the extension. Removing at most $i-1$ components from a nonincreasing list leaves stars that contain its suffix $n_i, \dots, n_s$. Hence the remaining graph contains a red $F_1^{(i)}$, a contradiction.

Every new blue edge meets X_B. Discarding at most $j-1$ components of a hypothetical blue F_2 likewise leaves a blue $F_2^{(j)}$ in $H - X$, another contradiction. □

Lemma 3.2 (Bipartite star avoidance)**.** *If a bipartite graph J satisfies $\Delta(J) \le a+b-2$, then it admits a coloring with no red $K_{1,a}$ and no blue $K_{1,b}$.*

Proof. A bipartite graph of maximum degree D decomposes into D matchings. For completeness, add isolated vertices to equalize the two part sizes, and then add edges, allowing parallel edges, until every vertex has degree D. The degree deficits on the two sides have equal sum, so this completion is possible. When $D > 0$, Hall's condition follows by counting the $D|S|$ edges from any set S in one part; it gives a perfect matching. Remove one perfect matching at a time to obtain a decomposition into D matchings. Restrict these matchings to J. The case $D = 0$ is immediate.

Color at most $a-1$ matching classes red and the remaining classes blue. Since $D \le a+b-2$, at most $b-1$ classes need be blue. The red maximum degree is at most $a-1$, and the blue maximum degree is at most $b-1$. □

Theorem 3.3 (Exact value with bipartite hosts)**.** *Every bipartite graph H with $H \longrightarrow (F_1, F_2)$ has at least L edges. Consequently the minimum number of edges among bipartite arrowing hosts is exactly L, for all positive lists and all net unit excesses.*

Proof. Put $M = s + t - 1$. Starting with $H_0 = H$, choose a vertex v_{q+1} of maximum degree in H_q and set $H_{q+1} = H_q - v_{q+1}$, for $0 \le q \le M-1$.

Fix such a q. For every feasible $i + j = q + 2$, Lemma 3.1 gives

$$H_q \longrightarrow (F_1^{(i)}, F_2^{(j)}).$$

In particular $H_q \longrightarrow (K_{1,n_i}, K_{1,m_j})$, because those are components of the two suffix targets. Since H_q is bipartite, Lemma 3.2 forces

$$\Delta(H_q) \ge n_i + m_j - 1.$$

Taking the maximum over the feasible pairs gives $\Delta(H_q) \ge \ell_{q+2} \ge 1$, so the chosen vertex exists at every step. Each edge is removed at most once, and therefore

$$e(H) \ge \sum_{q=0}^{M-1} d_{H_q}(v_{q+1}) = \sum_{q=0}^{M-1} \Delta(H_q) \ge \sum_{k=2}^{s+t} \ell_k = L.$$

The bipartite host H_* in Theorem 2.2 attains the bound. □

Remark 3.4 (Why this does not cover arbitrary hosts)**.** The implication $J \longrightarrow (K_{1,a}, K_{1,b}) \Rightarrow \Delta(J) \ge a+b-1$ is false for nonbipartite graphs. The triangle satisfies $K_3 \longrightarrow (P_3, P_3)$ but has maximum degree $2 < 3$. This is not a counterexample to the diagonal conjecture: the triangle has exactly the predicted three edges. It is a counterexample to the local degree implication needed by the bipartite proof.

4 The exact diagonal increment of a unit component

Write $F_1^+ = F_1 \sqcup K_2$ and define

$$\delta_1(F_1, F_2) = L(F_1^+, F_2) - L(F_1, F_2).$$

The analogous quantity δ_2 is defined by adjoining a unit to F_2.

Lemma 4.1 (Unit increment)**.** *For the original diagonal maxima ℓ_k,*

$$\delta_1(F_1, F_2) = m_t + \sum_{j=1}^{t-1} (m_j - \ell_{s+j+1})_+ . \tag{2}$$

In particular $m_t \le \delta_1(F_1, F_2) \le m_1$.

Proof. Appending a final entry 1 to the red list introduces a candidate of value m_j on diagonal $s + j + 1$. For $1 \le j \le t - 1$, this changes an existing maximum by $(m_j - \ell_{s+j+1})_+$. The final new diagonal contributes m_t, giving (2).

The lower bound is immediate. The old diagonal $s + j + 1$ contains the pair $(s, j + 1)$, so

$$\ell_{s+j+1} \ge n_s + m_{j+1} - 1 \ge m_{j+1}.$$

Thus each summand is at most $m_j - m_{j+1}$, and telescoping gives the upper bound $m_t + (m_1 - m_t) = m_1$. □

5 A conditional proof of the unrestricted identity

The following two statements are assumptions in this section. In particular, the second is not being asserted as an established lemma.

Hypothesis 5.1 (B: assumed base theorem)**.** The identity $\widehat{R}(F_1, F_2) = L(F_1, F_2)$ holds whenever $c \le 10$.

This is the statement of Theorem 11.1 in [1], used as an assumption at the user's request rather than independently verified.

Hypothesis 5.2 (U: unproved unit extension)**.** For every pair of nonempty star forests,

$$\widehat{R}(F_1 \sqcup K_2, F_2) \ge \widehat{R}(F_1, F_2) + \delta_1(F_1, F_2),$$
$$\widehat{R}(F_1, F_2 \sqcup K_2) \ge \widehat{R}(F_1, F_2) + \delta_2(F_1, F_2).$$

Theorem 5.3 (Conditional reduction)**.** *Hypotheses B and U imply the unrestricted identity* (1)*.*

Proof. For $i = 1, 2$, let u_i be the number of components K_2 in F_i. Then

$$c_i = u_i - \sum_{\substack{\text{component sizes } d \\ d \ge 3}} (d - 2) \le u_i.$$

Put $a = (c_1 - 10)_+$ and $b = (c_2 - 10)_+$. Delete a red-side unit components and b blue-side unit components, obtaining F_1^0, F_2^0. The deletions are possible by $c_i \le u_i$. The resulting forests are nonempty: if a side is trimmed, its new component count is at least ten, since $c_1 \le s$ and $c_2 \le t$.

Each unit deletion reduces the corresponding net unit excess by exactly one. Thus the two new excesses are $\min(c_1, 10)$ and $\min(c_2, 10)$. Hypothesis B gives

$$\widehat{R}(F_1^0, F_2^0) = L(F_1^0, F_2^0) =: L_0.$$

Restore the deleted units one at a time. Let $\mathcal{F}^{(0)}, \ldots, \mathcal{F}^{(a+b)}$ denote the successive ordered pairs, and let R_j and L_j denote their size Ramsey numbers and diagonal sums. At every restoration, Hypothesis U yields

$$R_{j+1} - R_j \geq L_{j+1} - L_j.$$

Summing gives

$$\widehat{R}(F_1, F_2) = R_{a+b} \geq R_0 + L_{a+b} - L_0 = L(F_1, F_2).$$

The universal upper bound supplies the reverse inequality. □

Remark 5.4 (Logical strength of U)**.** If the unrestricted identity already holds, both inequalities in U hold with equality by the definition of δ_i. Together with Theorem 5.3, this means that, under B, U is equivalent to the unrestricted identity. Merely stating U therefore does not supply the missing proof.

Corollary 5.5 (A quantitative consequence of B alone)**.** *Assuming B, but not U, every pair satisfies*

$$L - m_1 (c_1 - 10)_+ - n_1 (c_2 - 10)_+ \leq \widehat{R}(F_1, F_2) \leq L.$$

More precisely, the trimmed pair in the preceding proof gives the lower bound $\widehat{R}(F_1, F_2) \geq L_0$.

Proof. Every host arrowing (F_1, F_2) also arrows (F_1^0, F_2^0), so B implies $\widehat{R}(F_1, F_2) \geq L_0$. Restore first the a red units and then the b blue units. By Lemma 4.1, each red restoration increases the diagonal sum by at most m_1, and each blue restoration by at most n_1. Restoring units does not increase either head. Thus $L - L_0 \leq am_1 + bn_1$. Apply Theorem 2.2 for the upper bound. □

6 A concrete proof issue in the supplied manuscript

The proof of Theorem 5.1 in [1], in its $s = 1$ case, asserts that deleting a vertex of degree at least $b + 1$ from a host arrowing (P_3, F_2) leaves a host arrowing (K_2, F_2). This implication fails.

Take $F_2 = K_{1,4}$ and $H = K_{1,5}$. Every coloring of H has either at least two red edges, giving a red P_3, or at least four blue edges, giving a blue $K_{1,4}$. Hence

$$H \longrightarrow (P_3, K_{1,4}).$$

Its center v has degree five, but $H - v$ is edgeless and does not arrow $(K_2, K_{1,4})$. This refutes that auxiliary implication, not the claimed numerical theorem. It explains why Hypothesis B is not being silently upgraded from an assumption to a verified result.

Section 12 of [1] also explicitly states that no general excess-k to excess-$(k + 1)$ lemma has been established there. The unrestricted lower bound remains unproved in this note. The complete unconditional statements here are Theorems 2.2 and 3.3, together with the elementary lemmas supporting them.

References

[1] *The size Ramsey number of two star forests when the net unit excess is at most 10.* User-supplied manuscript, dated 29 September 2026, supplied as `size_ramsey_net_ten.pdf` and `size_ramsey_net_ten.tex`. In particular, Sections 3, 5, 11 and 12. Unpublished; the claimed base theorem is assumed, not independently certified in this note.

[2] Pingting Fu, Zhidan Luo and Zhenyu Ni. *Size Ramsey minimal graphs for uniform star forests.* arXiv:2606.04439v3, 4 July 2026. `https://arxiv.org/abs/2606.04439v3`.

[3] Akbar Davoodi, Ramin Javadi, Azam Kamranian and Ghaffar Raeisi. *On a Conjecture of Erdős on Size Ramsey Number of Star Forests.* arXiv:2111.02065, 2021. `https://arxiv.org/abs/2111.02065`.